

Jun Hong

PERSONAL INFORMATION

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U.S. Citizen

Mathematician and university lecturer transitioning to data science and scientific machine learning.

EDUCATION

B.S. Statistics & Data Science, Physics, and Chemistry

University of Colorado, Boulder

Expected May 2029

Undergraduate GPA: 3.67

Planned: intra-university transfer to add a Computer Science B.S., January 2027

In progress (Fall 2026):

- Software Development Methods and Tools
- Quantum Mechanics 2
- Introduction to Data Science
- Partial Differential and Integral Equations (graduate)
- Molecular Physics (graduate)
- Experimental Physics 2
- Physical Chemistry Lab 1

Relevant completed coursework: Graduate Probability, Mathematical Statistics, Numerical Computation, Computer Systems, Data Structures, Computer Graphics, Physical Chemistry, Electricity and Magnetism

M.A. Mathematics, December 2018

University of Colorado, Boulder

Graduate GPA: 3.74

B.A. Mathematics, December 2013

University of California, Berkeley

Undergraduate GPA: 3.20

EMPLOYMENT

Department of Mathematics, University of Colorado, Boulder

Lecturer

August 2020 – May 2026

- Instructor of record for Statistics, Calculus, and Linear Algebra — 4 sections per term, 30 students each.
- Taught 1,500+ students across 17 semesters, with responsibility for lectures, office hours, and grading.

Primary Instructor / TA

August 2014 – May 2020

- Instructor for introductory mathematics courses as a graduate student; earlier led recitations and graded as a teaching assistant.

PROJECTS

Computer Graphics

github.com/j01001000/computer-graphics

- Rendered Miller's planet from *Interstellar* — cube-mapped environment, lighting from surface normals, and a 2D wave simulation for its giant wave
- Visualized hydrogen-atom orbitals by sampling analytic wavefunctions into instanced GPU particle systems
- Simulated gravitational lensing (an Einstein ring) around a non-rotating black hole by integrating photon paths with RK4 in OpenCL over a sphere-mapped background

RESEARCH EXPERIENCE

Random Walks on Unipotent Groups

May 2015 – July 2015

- Computed transition probabilities for random walks on unipotent matrix groups — how right-multiplication by a matrix of shifted rank 0 or 1 moves elements between superclasses.

AWARDS

Continuing Education Nontraditional Student Scholarship, Fall 2023.